The future of AI based E2E test automation

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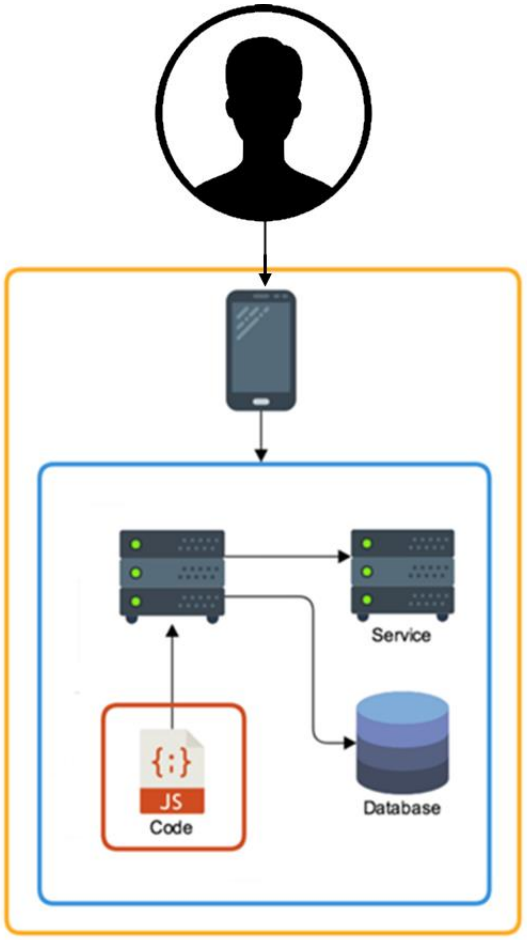
applitools
AI Powered Testing & Monitoring

Agenda

- Why E2E testing?
- Challenges with automated E2E testing
- How can AI assist today?
- What role will AI play (in the very near) future?
- Q&A

E2E (UI based) test automation

- Verify the correctness of a system through its UI
  - Simulating user interactions as input
  - Asserting the output rendered in the UI
- Supported by dozens of tools and frameworks
  - Selenium, Cypress, Playwright, WDIO, Appium, …

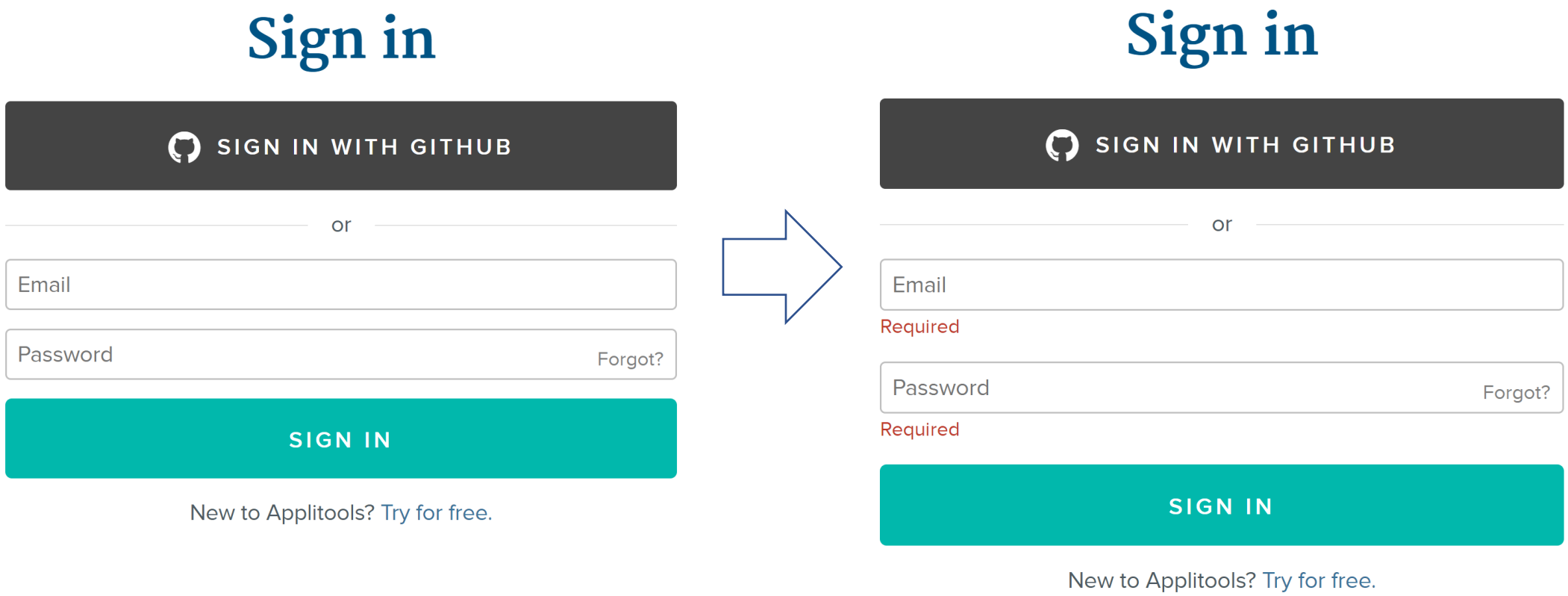




E2E testing is critically important!

What’s wrong with traditional E2E test automation?

Example UI test: required login details



A traditional UI test

```
1   describe('Testing our login screen', () => {
2     it('properly warns on missing input', () => {
3       cy.visit('https://applitools.com/users/login');
4
5       cy.get('button[type=submit]')
6         .click();
7
8       cy.get('.validation-error')
9         .eq(0)
10        .should('contain', 'Required');
11
12      cy.get('.validation-error')
13        .eq(1)
14        .should('contain', 'Required');
15
16      cy.get('button[type=submit]')
17        .should('contain', 'Sign in');
18    });
19  });
```



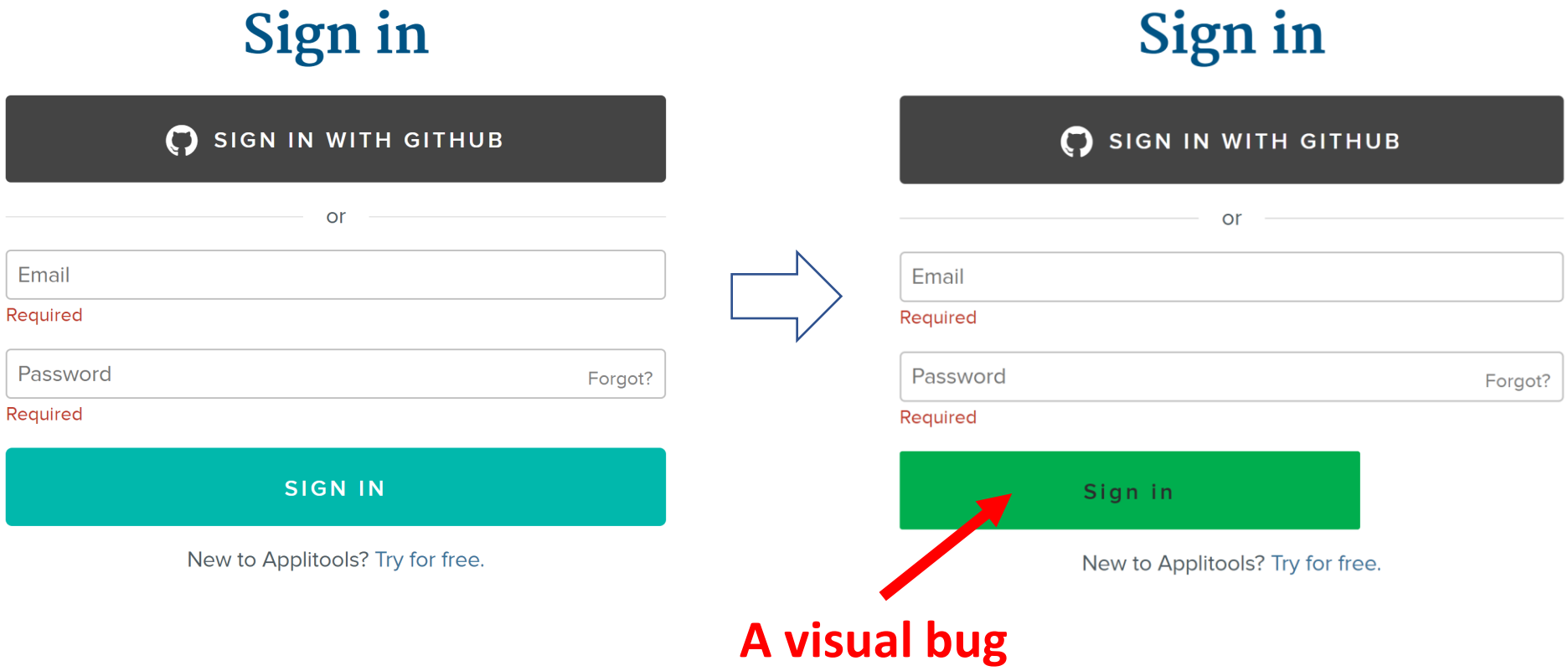
Need to write a lot of code

Code will break when the UI changes

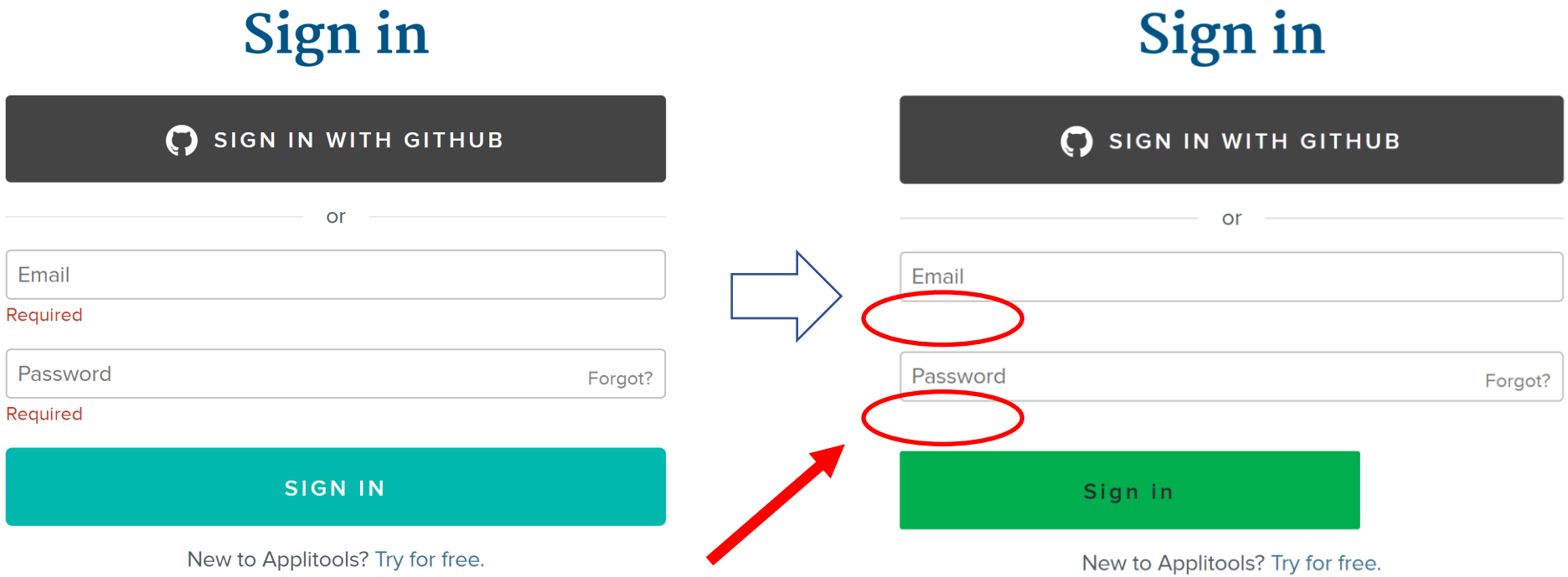
When the UI changes…

```
08:31:17        Info    Selection       103     0       "[Installation Location: {Installation location:=C:\Program Files\SolidWorks Corp (Ena
08:31:17        Info    Selection       105     0       "Estimated additional installation size: 656 MB; Estimated download size: 386 MB"
08:33:17        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:35:10        Info    Selection       37      55611   "The user clicked the button representing the {0: <U>N</U>ext} operation on the {1: Su
08:35:10        Info    Selection       39      55613   "The following lines show the selections for the Summary page when the user selected t
08:35:10        Info    Selection       99      0       "You are updating 2013 from SP 1.0 to SP 4.0."
08:35:10        Info    Selection       102     0       "[Download Options: {Operation:=Download and install (Enabled: 1)}, {Download to:=\\SE
08:35:10        Info    Selection       103     0       "[Installation Location: {Installation location:=C:\Program Files\SolidWorks Corp (Ena
08:35:10        Info    Selection       105     0       "Estimated additional installation size: 656 MB; Estimated download size: 386 MB"
08:35:48        Info    Status  166     0       "DOWNLOAD SPACE REQUIRED: {0: 641750}, AVAILABLE: {1: 783667496}, folder: {2: \\SERVERNAME\Eng
08:36:10        Error   Message 9       55200   "Dialog Shown: {0: Toolbox is currently in use. Make sure that no one is using Toolbox then cl
08:36:10        Info    Selection       43      55617   "The source folder that will be used is: {0: \\SERVERNAME\Engineering\Users\USERNAME\S
08:36:10        Info    Selection       44      55617   "The Installation Manager Source Folder that will be used is: {0: \\SERVERNAME\Enginee
08:36:10        Info    Step    25      55502   "User advanced from '{0: Summary}' page to '{1: Download Progress}' page."
08:36:18        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:38:55        Info    Step    25      55502   "User advanced from '{0: Download Progress}' page to '{1: Countdown}' page."
08:39:16        Info    Step    25      55502   "User advanced from '{0: Countdown}' page to '{1: Install Progress}' page."
08:39:16        Info    Selection       107     0       "Writing serial numbers to the registry..."
08:39:16        Warning Selection       110     0       "===> CHANGING {0: solidworks} SERIAL NUMBER IN REGISTRY before installs: {1: XXXXXXXX
08:39:19        Warning Selection       110     0       "===> CHANGING {0: cae_cwcore} SERIAL NUMBER IN REGISTRY before installs: {1: XXXXXXXX
08:39:20        Warning Selection       110     0       "===> CHANGING {0: cae_cosmosfloworks} SERIAL NUMBER IN REGISTRY before installs: {1:
08:39:21        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:39:21        Info    Status  138     0       "CREATING INSTALL STEP 0 for SLDPS: cost=4096KB"
08:39:21        Info    Status  143     0       "CREATING INSTALL STEP 2 to Install: prod=CheckForUpdates, code=, upg=, cost=15061KB, force=0,
08:39:21        Info    Status  143     0       "CREATING INSTALL STEP 3 to Install: prod=Microsoft_VBA_KB2783832, code=, upg=, cost=0KB, forc
08:39:21        Info    Status  143     0       "CREATING INSTALL STEP 4 to Upgrade: prod=SolidWorks, code={B6B5EA7E-B91F-443D-A958-B0062FB538
08:39:22        Info    Status  143     0       "CREATING INSTALL STEP 5 to Upgrade: prod=eDrawings, code={2686CB6D-19AF-49A0-ABD2-29D134D7C23
08:39:22        Info    Status  143     0       "CREATING INSTALL STEP 6 to Upgrade: prod=Explorer, code={168EB20E-FC09-4D2E-83A9-49483710304C
08:39:22        Info    Status  143     0       "CREATING INSTALL STEP 7 to Upgrade: prod=COSMOSFloWorks, code={2A6228B3-BB5D-47D9-9BB6-20953A
08:39:22        Info    Status  143     0       "CREATING INSTALL STEP 8 to Upgrade: prod=SolidWorksPlastics, code={BA812540-2D88-4A6A-A527-E7
08:39:41        Info    Status  139     0       "CREATING INSTALL STEP 9 to FINISH: cmd=, cost=72348KB"
08:40:03        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:40:21        Info    Step    92      0       "{0: Installing} EXE: {1: \\SERVERNAME\Engineering\Users\USERNAME\SolidWorks Downloads\Solidwo
(ID {3: CheckForUpdates}, lang {4: }, pcode {5: }, mode {6: 3}, upg = {7: }, f{8: 0}, costKB {9: 15061})"
08:40:24        Info    Step    94      0       "** INSTALLATION SUCCEEDED **"
08:40:38        Info    Step    91      0       "{0: Installing} MSP: {1: \\SERVERNAME\Engineering\Users\USERNAME\SolidWorks Downloads\Solidwo
(ID {4: Microsoft_VBA_KB2783832}, lang {5: }, pcode {6: }, mode {7: 3}, upg = {8: }, f{9: 0}, costKB {10: 0})"
08:40:41        Info    Step    155     0       "** Patch SUCCEEDED **"
08:40:53        Info    Step    91      0       "{0: Installing} MSP: {1: \\SERVERNAME\Engineering\Users\USERNAME\SolidWorks Downloads\Solidwo
(ID {4: SolidWorks}, lang {5: }, pcode {6: {B6B5EA7E-B91F-443D-A958-B0062FB53804}}, mode {7: 3}, upg = {8: 20130-40100-1100}, f{9: 0}, costKB
08:42:20        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:45:18        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:48:17        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:51:16        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:54:16        Info    Status  58      55803   "Received data from Installation Manager server (Script: {0: http://im.solidworks.com/bin2013/
08:54:22        Error   Status  156     0       "***START DUMP OF WINDOWS INSTALLER DATA***"
08:54:22        Error   Status  157     0       "=== Logging started: 7/19/2013  8:40:57 ==="
08:54:22        Error   Status  157     0       "Action start 8:40:57: INSTALL."
08:54:22        Error   Status  157     0       "Action start 8:40:57: ProgramFilesFolder.25459867_D5E7_32F7_9C7A_5EFF02A207F6."
```

Can’t catch unexpected visual bugs!



Can’t catch unexpected functional bugs!



Automated E2E testing challenges

- Thousands of tests required to cover a modern app
- Low coverage
- High maintenance overhead
- Skill set dependent

How can AI assist with E2E testing today?

How can AI assist with E2E testing today?

- Replace your traditional assertions with visual validations
- Stabilize your builds with self-healing locators

Replace assertions with visual checkpoints

```
describe('Testing our login screen', () => {
  it('properly warns on missing input', () => {
    cy.visit('https://applitools.com/users/login');

    cy.get('button[type=submit]')
      .click();

    cy.get('.validation-error')
      .eq(0)
      .should('contain', 'Required');

    cy.get('.validation-error')
      .eq(1)
      .should('contain', 'Required');

    cy.get('button[type=submit]')
      .should('contain', 'Sign in');
  });
});
```

Test results

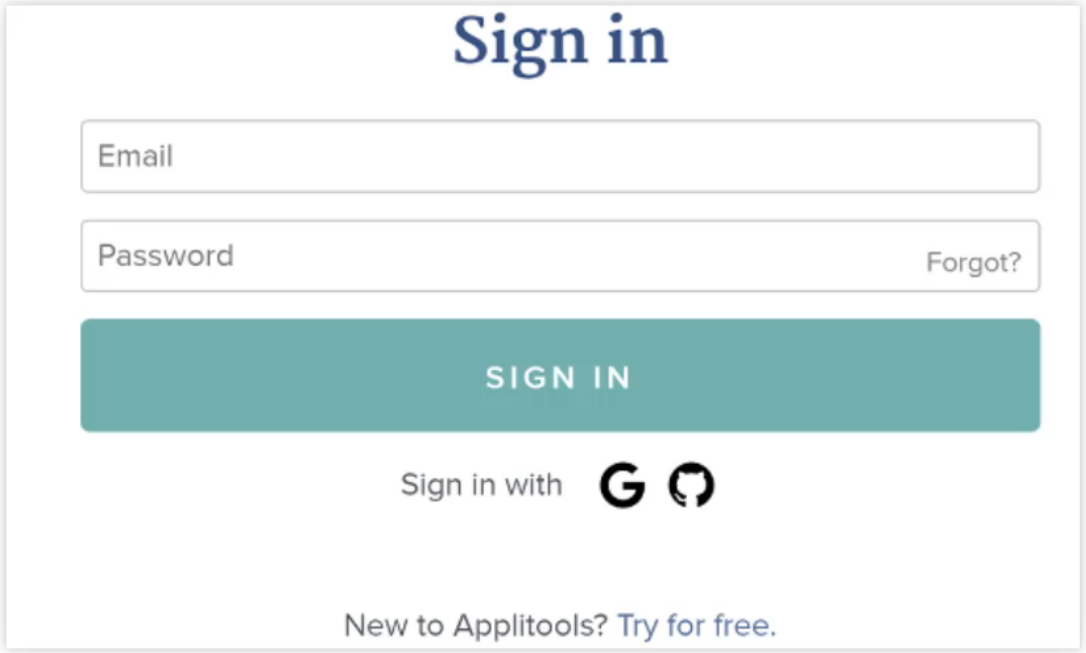
Test: Test login

Steps: 1 | Unresolved: 1

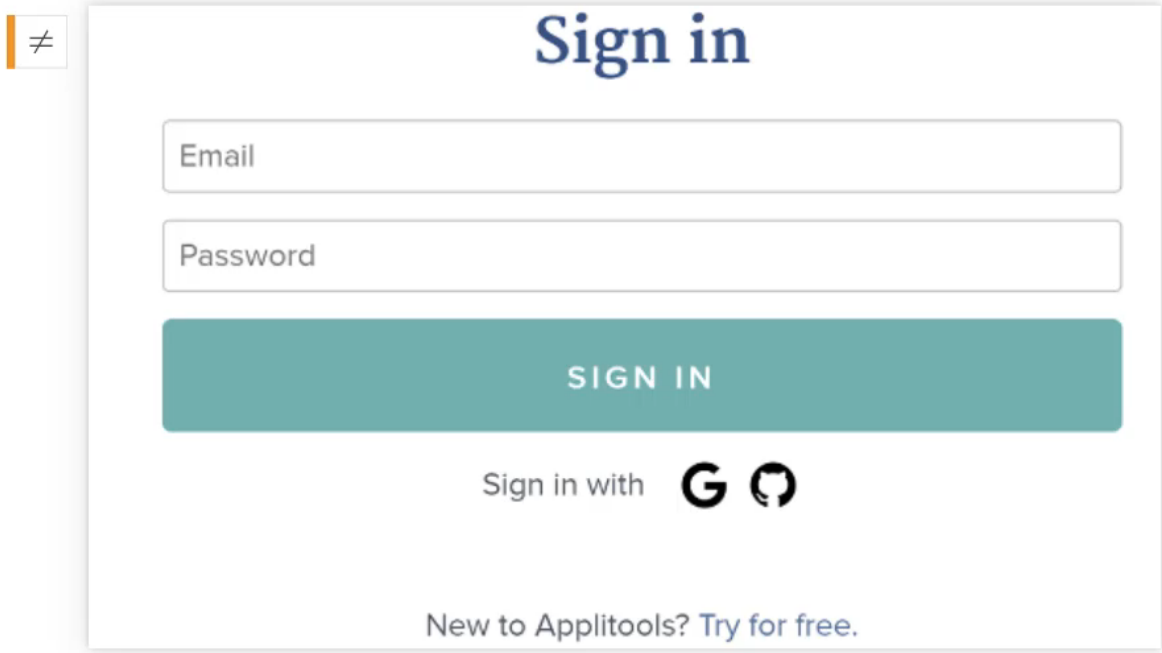
VIEW | HIGHLIGHT DIFFS | ANNOTATIONS | AUTO MAINTENANCE | ACCESSIBILITY

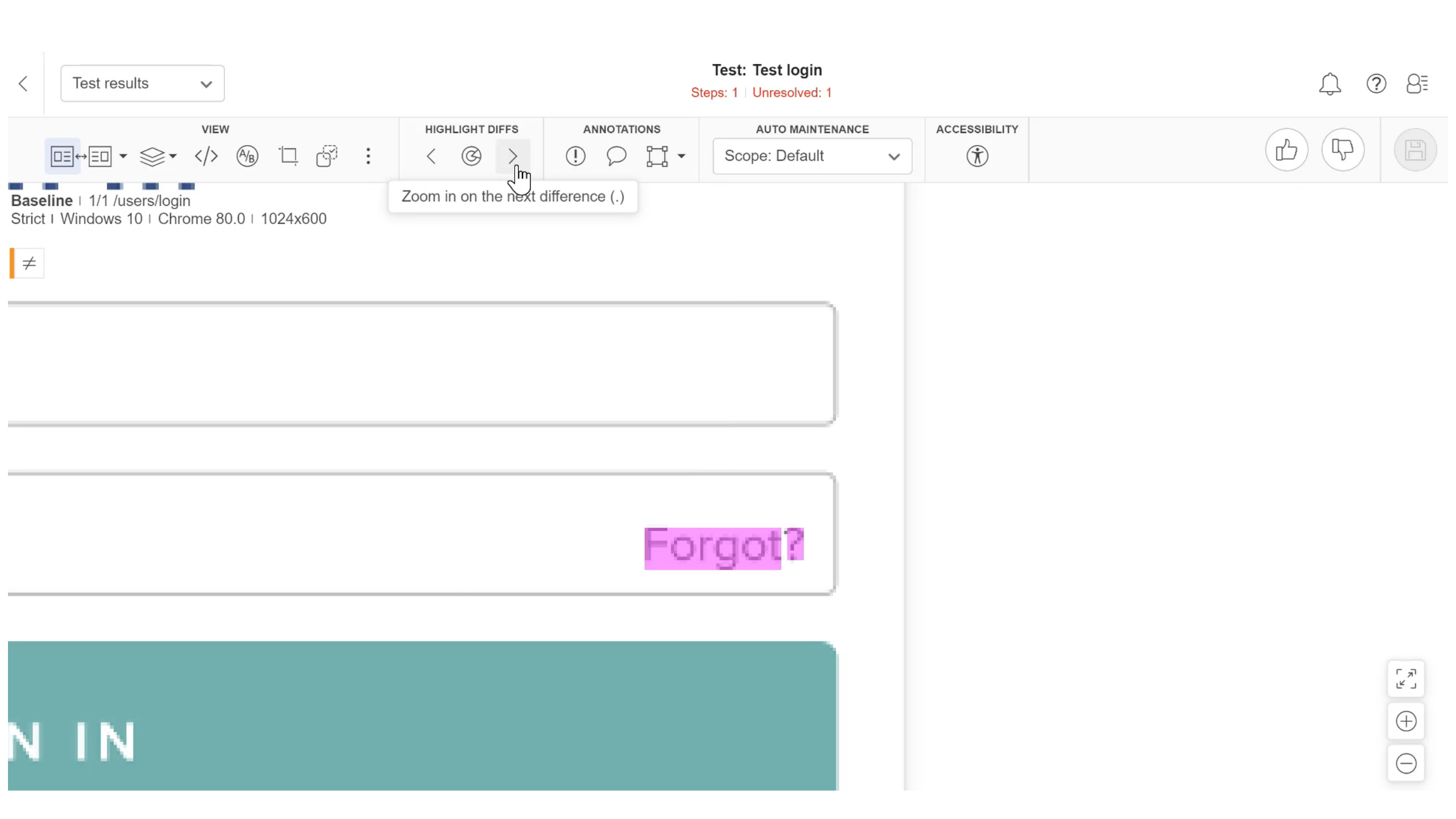
Scope: Default

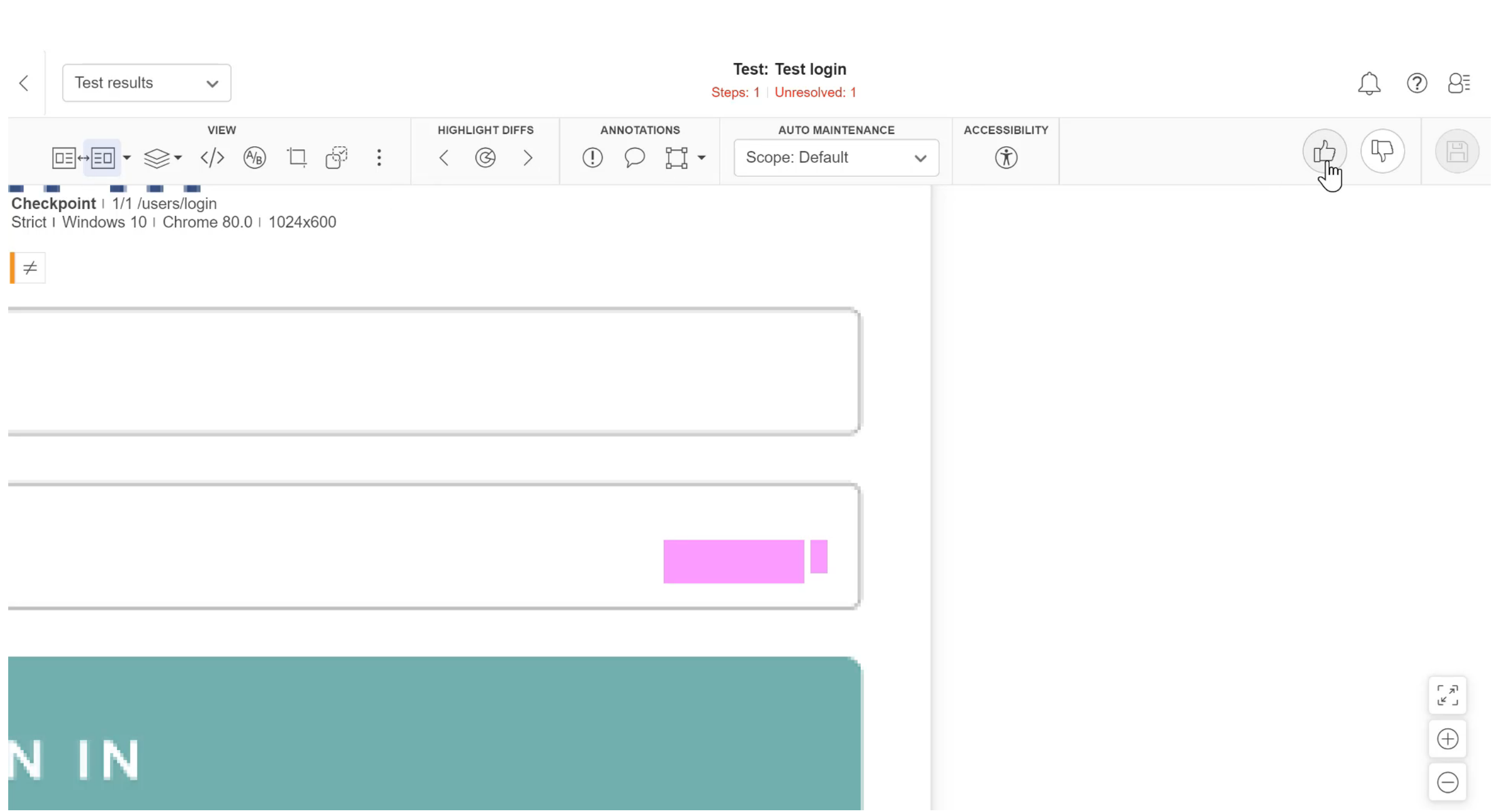
Baseline | 1/1 /users/login
Strict | Windows 10 | Chrome 80.0 | 1024x600



Checkpoint | 1/1 /users/login
Strict | Windows 10 | Chrome 80.0 | 1024x600




Test results
Test: Test login
Steps: 1 | Unresolved: 1
VIEW
HIGHLIGHT DIFFS
ANNOTATIONS
AUTO MAINTENANCE
Scope: Default
ACCESSIBILITY
Zoom in on the next difference (.)
Baseline | 1/1 /users/login
Strict | Windows 10 | Chrome 80.0 | 1024x600
≠
Forgot?
N IN


Test results
Test: Test login
Steps: 1 | Unresolved: 1
VIEW
HIGHLIGHT DIFFS
ANNOTATIONS
AUTO MAINTENANCE
Scope: Default
ACCESSIBILITY
Checkpoint | 1/1 /users/login
Strict | Windows 10 | Chrome 80.0 | 1024x600
≠
N IN

Visual checkpoints advantages

- Enhances existing tests and reduces the amount of test code by up to 80%
- A single visual checkpoint for complete visual and functional coverage
- Catches unexpected defects
- Does not break when the UI changes
- No coding skills required to maintain validation logic

```
describe('Testing our login screen', () => {
  it('properly warns on missing input', () => {
    cy.visit('https://applitools.com/users/login');

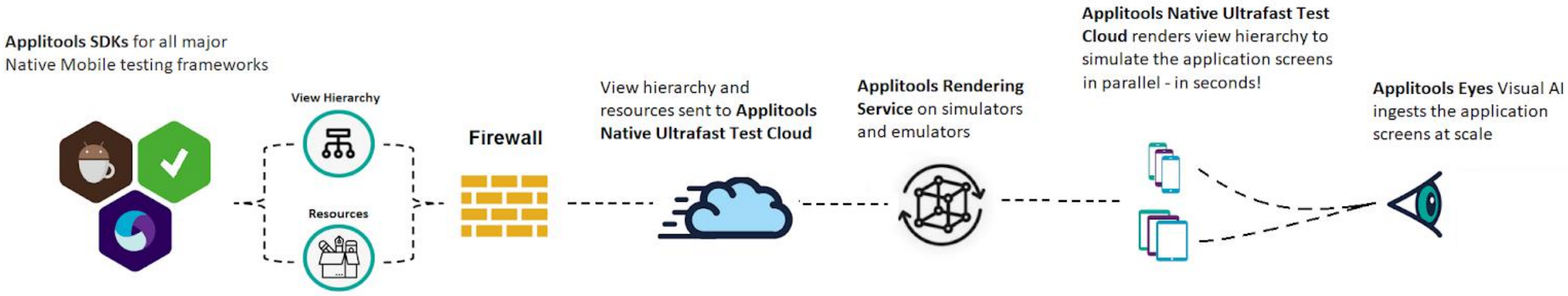
    cy.get('button[type=submit]')
      .click();

    cy.get('.validation-error')
      .eq(0)
      .should('contain', 'Required');

    cy.get('.validation-error')
      .eq(1)
      .should('contain', 'Required');

    cy.get('button[type=submit]')
      .should('contain', 'Sign in');
  });
});
```

Ultrafast cross device and browser testing



- ✓ Innovative
- ✓ Fast
- ✓ Stable
- ✓ Secure
- ✓ Simple

Demo

E2E testing with Cypress and Applitools Eyes

AI can stabilize your builds with self-healing locators

Element locators are the root of all evil

```
describe('Testing our login screen', () => {
  it('properly warns on missing input', () => {
    cy.visit('https://applitools.com/users/login');

    cy.get('button[type=submit]')
      .click();

    cy.get('.validation-error')
      .eq(0)
      .should('contain', 'Required');

    cy.get('.validation-error')
      .eq(1)
      .should('contain', 'Required');

    cy.get('button[type=submit]')
      .should('contain', 'Sign in');
  });
});
```

Element locators break following UI changes

Element locators breaking is the primary cause for **delayed test feedback**

and for **high test maintenance overhead** following UI changes

Self healing applies AI to **find an element even if its locator is broken**

using information learned from previous successful runs

How does self-healing work?

- Every time we find an element
  - Capture hundreds of data points about the element
    - All attributes, location in hierarchy, details of ancestor and neighbor elements
  - Store data in a DB using the locator as key
- When we can’t locate an element using a given locator
  - Retrieve information from the DB using the failed locator
  - Use proprietary algorithms to find the element based on that information
  - If successful, update the DB and suggest a new locator in our dashboard
  - Return the found element

What can be healed?

- Self-healing can find an element even if simultaneously
  - Element properties change (e.g., ID, class, tag name, custom, etc)
  - Text changes (clickable text, input value, label, placeholder)
  - DOM position changes (hierarchy, position in list)
  - Size and location changes
- Adaptive
- Implicitly wait for elements

The huge benefits of self-healing

- Tests don’t crash and don’t delay feedback to developers
  - Release faster and avoid dev context switching
- No need to fix broken locators following UI changes
  - Although we report them and suggest how to fix them
- Save hours reviewing failed test reports and fixing broken locators
  - Can double the efficiency of each team member

When is self healing useful?

- Avoid test failures and test maintenance following UI changes
- Poor locator authoring skills
- Apps with weak locators
- Apps with dynamically generated UI (dynamic ids)

Demo

Self healing locators with the Applitools Execution Cloud

How can AI assist with E2E testing today?

- Drastically increase test coverage
- Reduce the amount of test code to author and maintain
- Eliminate repetitive maintenance work
- Lower required skill set
- Improve build stability
- Shorten feedback loops and increase team velocity

What role will AI play in the future?

Autonomous testing platforms

- Create an entire test suite for your app out of the box
  - Automatic app exploration
  - Automatic validations (regressions, benchmarks, generic test flows)
  - Doesn’t have to provide 100% coverage to be valuable
- The more generic your app is, the better it will work
- Guide the AI where it needs help
  - Training
  - Test code
  - Record interactions with the UI
  - In plain English: “Click the link below the blue button at the bottom of the page”

How to evaluate an autonomous tool?

- What coverage are you getting?
  - Discovery, validation, coverage changes
- Ongoing maintenance overhead
  - How well will the tool adapt to application changes?
  - How does the tool facilitate large scale test maintenance?
- What skills are required to operate the tool?
  - Testing, coding, element locators, ML, Devops, etc.
- How easy is it to guide the AI where it needs help?

applitools
Autonomous

Home
Dashboard
Results
Plans
Test management
Test library
Configuration

Settings
Notifications
10

Team:
Ayelet's team
Moshes's team
+6 more

Rowan Matthews

Test library

Automatically create tests for your full website



Let's get started!

Enter URL of your website https://example.com/

Your test environments:

1440x900 1200x857 1080x1920

Advanced

Cancel Next

Demo

Prompt based test execution + ChatGPT

ChatGPT

Give me step by step instructions for ordering the Dune book in amazon.com for Kindle using my default payment details. Include only mandatory steps assuming I'm already logged in and in amazon.com. Each step should consist of a single line and should start with an action such as "click" or "type" and followed by a description of the target element. include only interaction steps and skip validation steps. Include only a single option in each step.

What role will testers play in the future?

What role will human testers play?

- Testing will become more important than ever
  - AI is not 100% accurate and so will the app code it will generate!
  - Less skilled developers
  - Much more (AI generated) code to test
- Human testers will need to
  - Decide on the testing strategy and tools
  - Guide and fine tune the AI
  - Determine if a change is a bug or a feature
  - Test everything the AI won’t be able to

Questions?

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